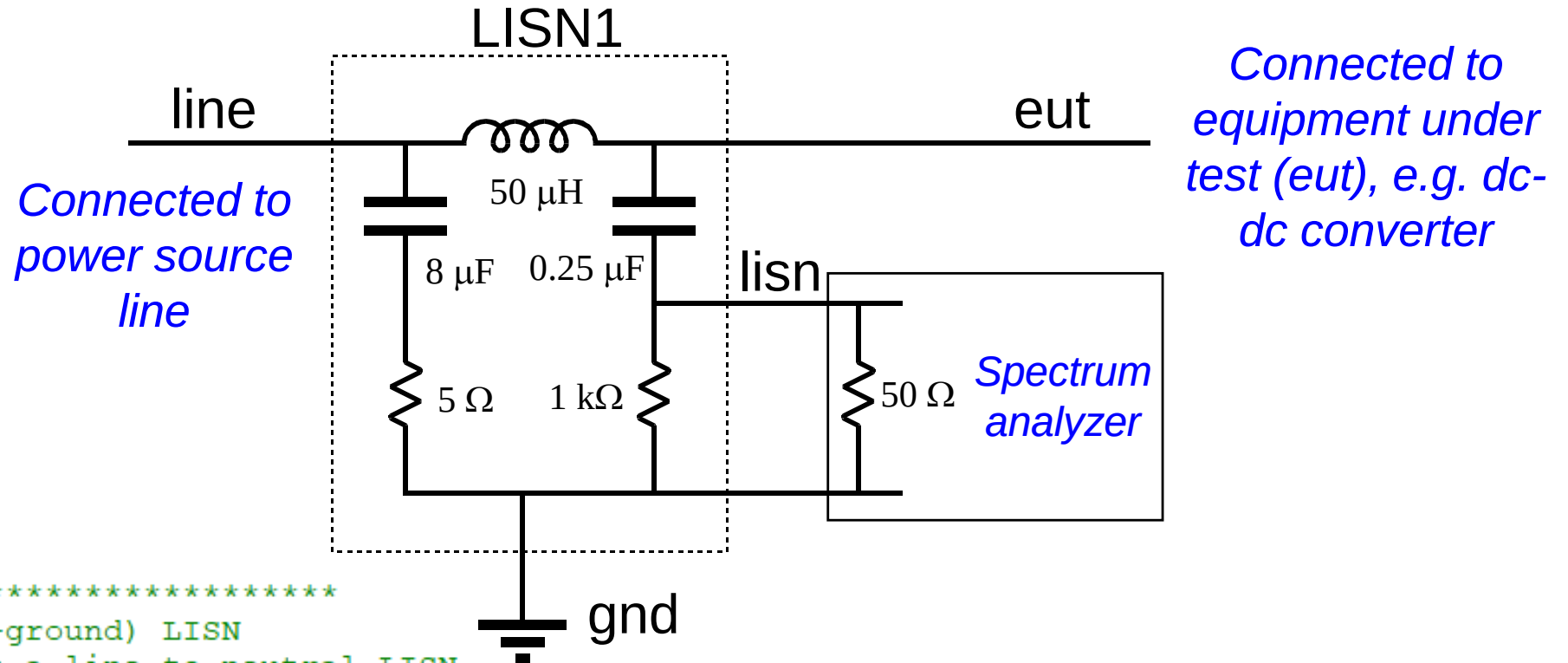


- Measurement of conducted EMI and the purpose of Line Impedance Stabilization Network (LISN)
- Differential-mode and common-mode EMI
- Simulation example: conducted EMI in a boost dc-dc converter

An example of 1/2 of the line impedance stabilization network

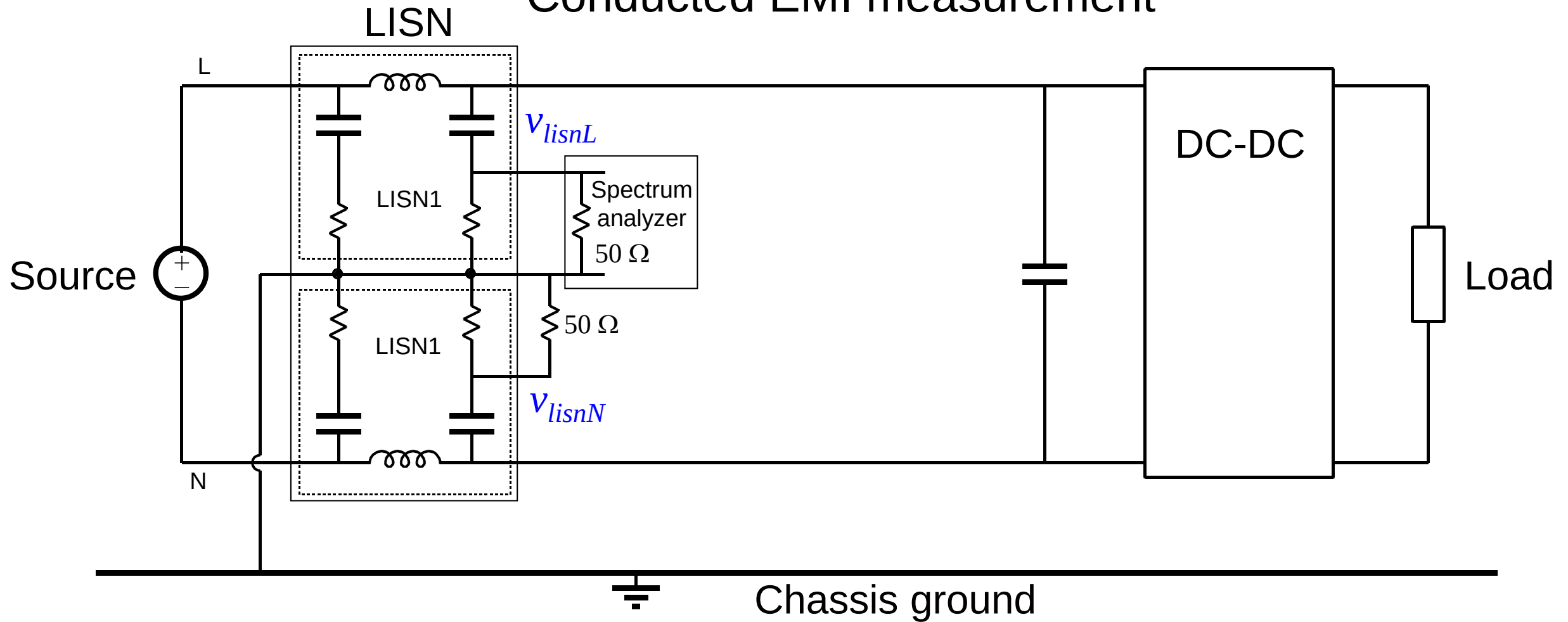


Spice sub-circuit netlist

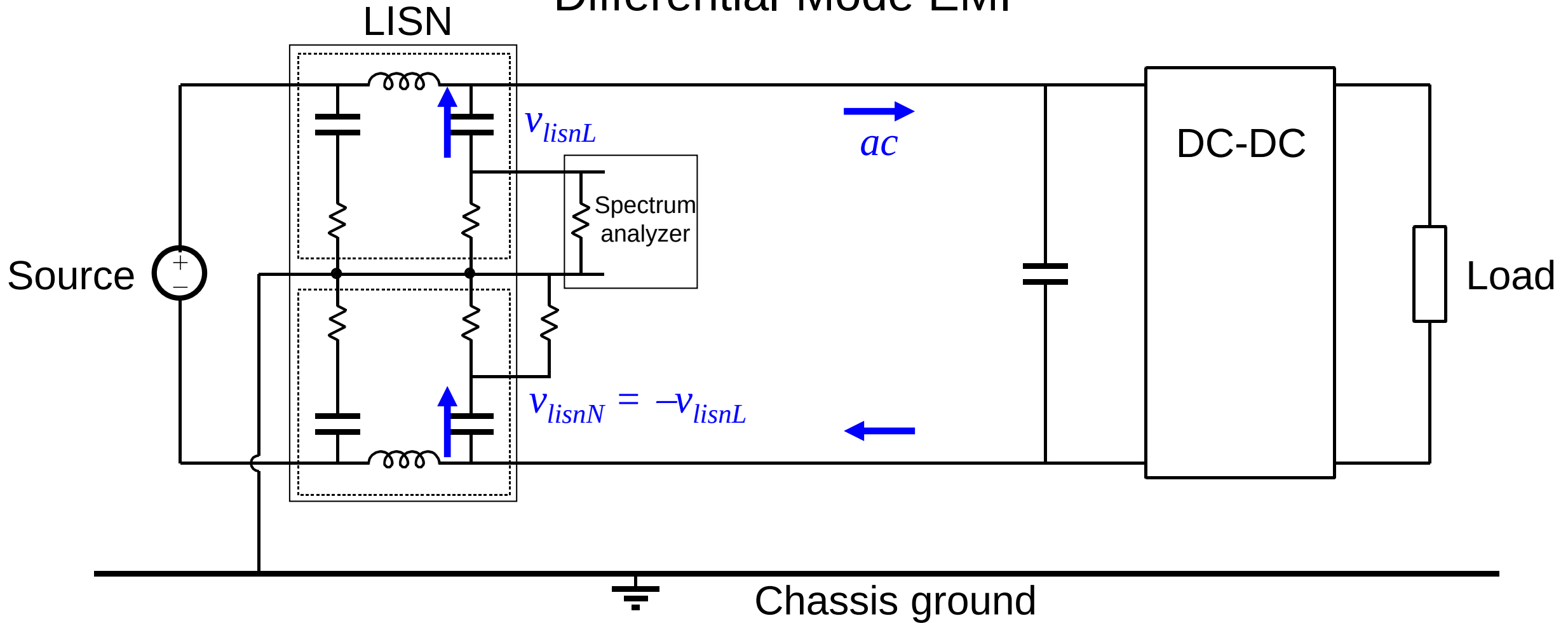
```
*****
* Single-channel (line-to-ground) LISN
* Connect two LISN1 to get a line-to-neutral LISN
*****
.subckt LISN1 line eut lisn gnd
*
L1 1 dut 50uH
R1 line 1 10m
Cline line 2 8uF
Rline 2 gnd 5
Clisn dut lisn 0.25uF
Rlisn lisn gnd 1k
.end
```

The purpose of a LISN is to standardize the test setup for EMC compliance testing

Conducted EMI measurement

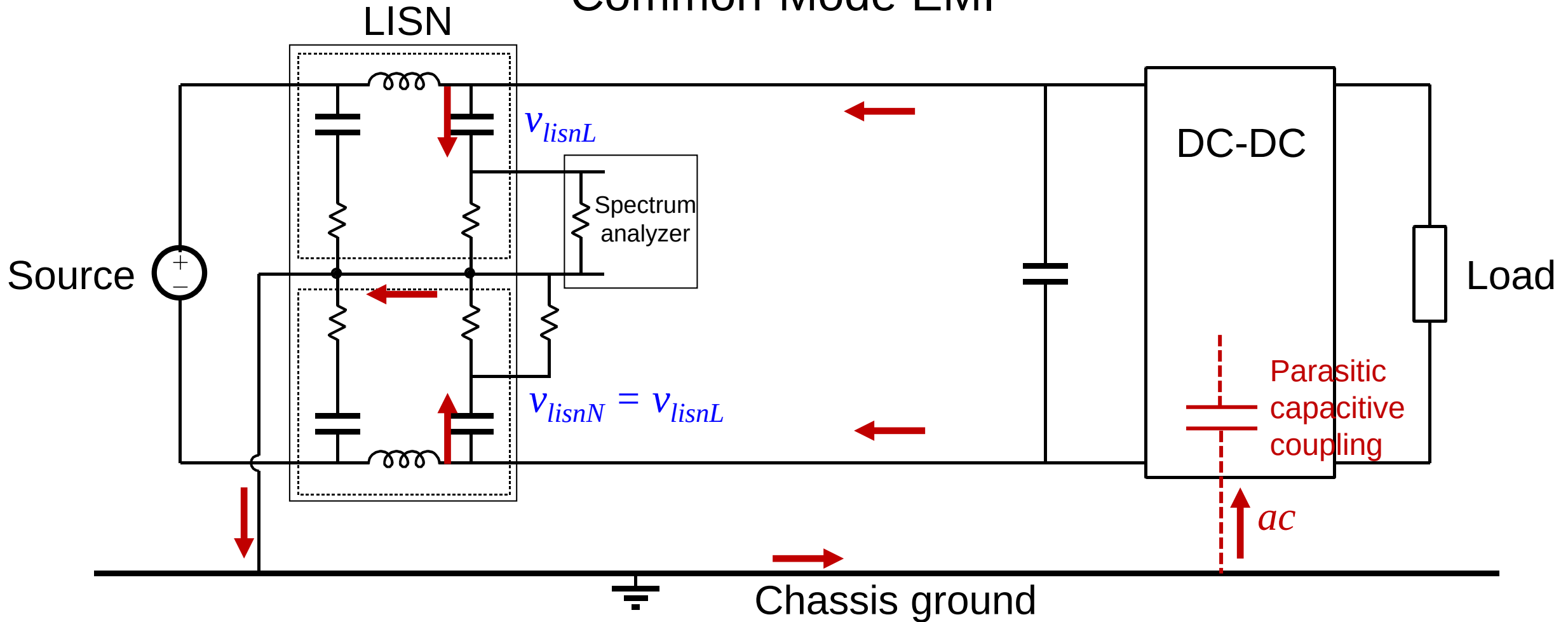


Differential-Mode EMI



Differential-Mode EMI is determined by the converter input current ripple, which depends on converter configuration, parameters and switching frequency

Common-Mode EMI



Common-Mode EMI is determined by currents through parasitic capacitances between high dv/dt nodes and chassis ground (such as from transistor drain to heat sink). Common-mode EMI depends on dv/dt , component parasitics, circuit and mechanical layout.

Electromagnetic compatibility (EMC) standards

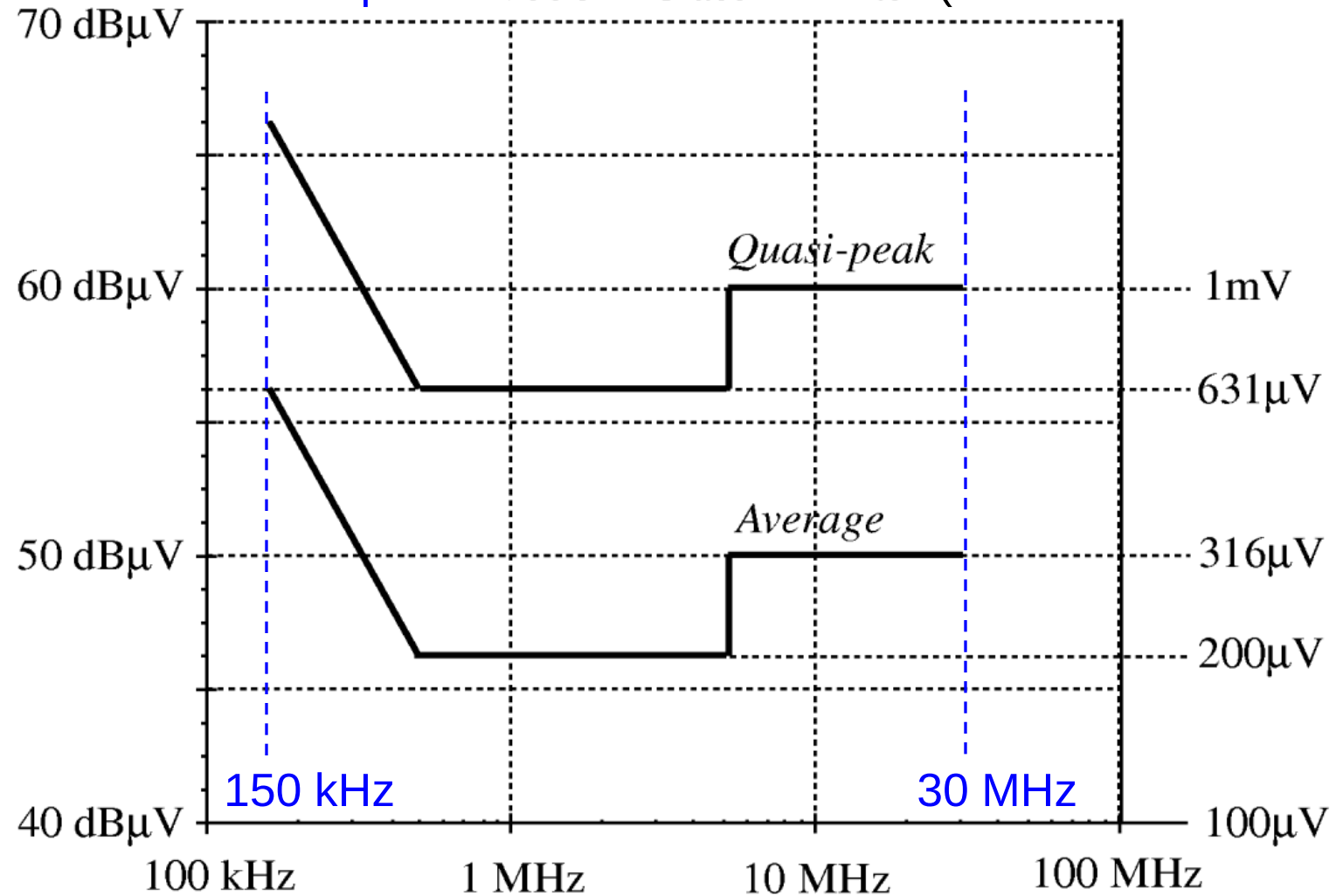
Example: EN 55022 Class B limits (also known as CISPR 22)

Both:

$$20\log\left(\frac{\|v_{lisnL}\|}{1\ \mu\text{V}}\right)$$

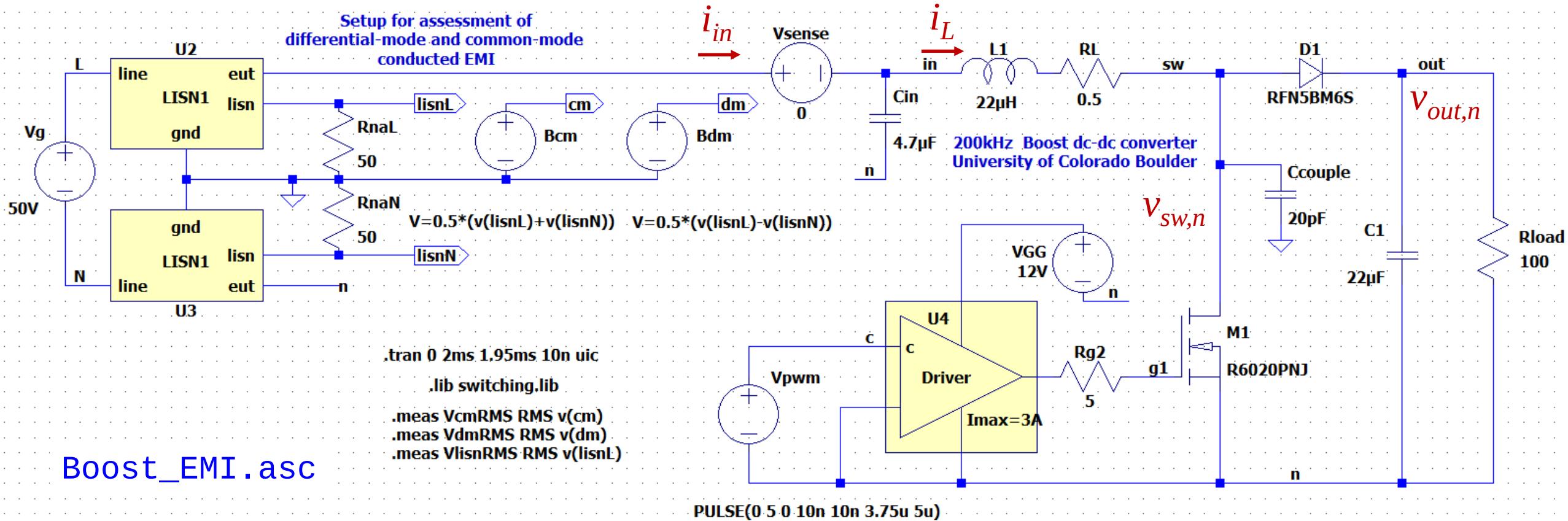
$$20\log\left(\frac{\|v_{lisnN}\|}{1\ \mu\text{V}}\right)$$

measured by the
spectrum analyzer
must be below the
limits



Note: “Quasi-peak” and “Average” are two different measurement settings on the spectrum analyzer

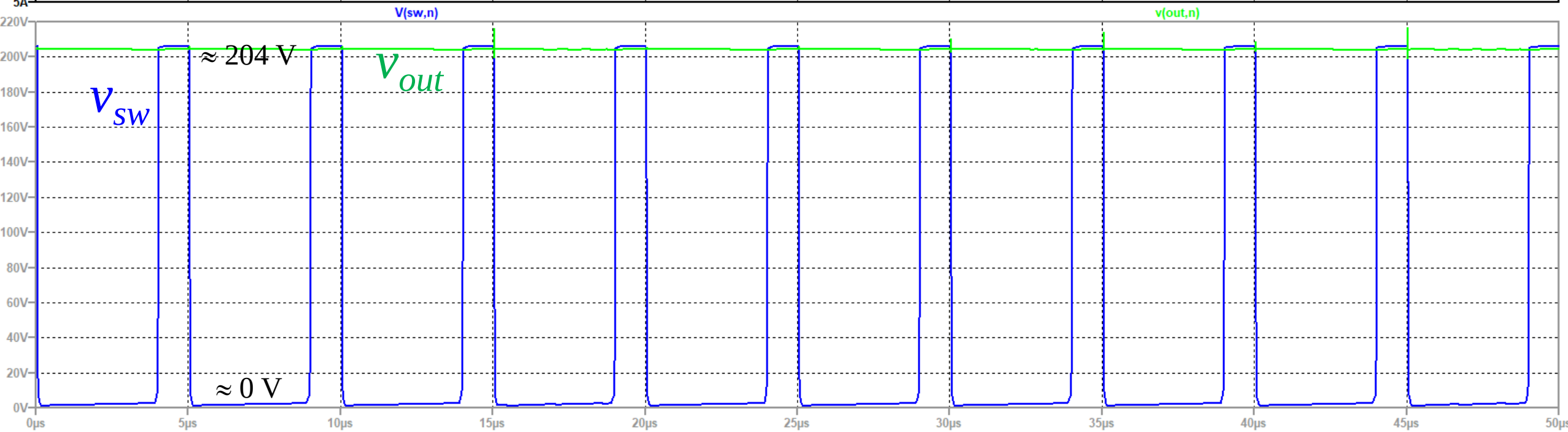
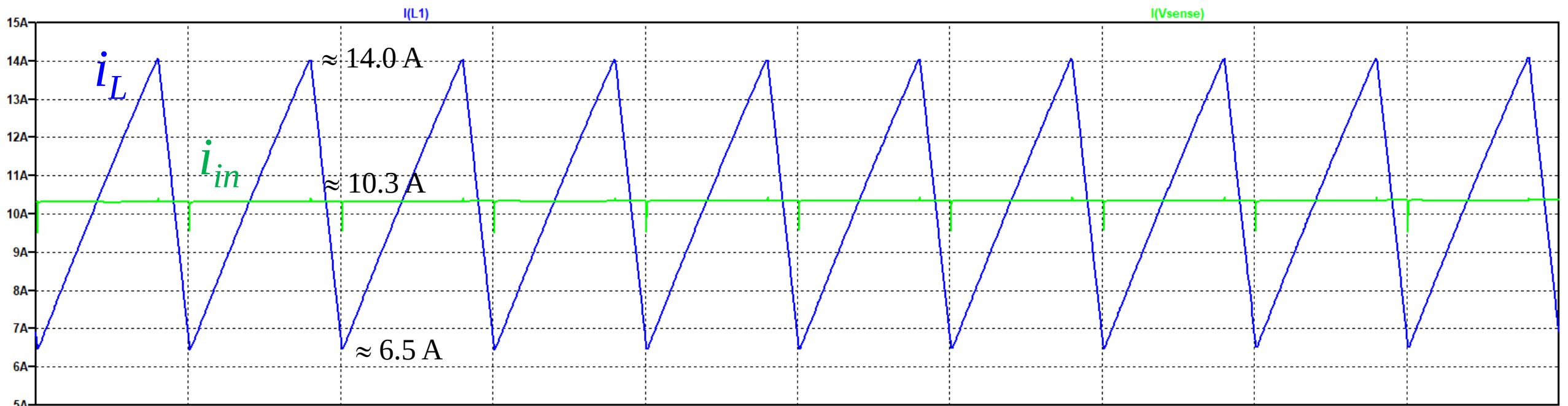
Conducted EMI in a Boost DC-DC Converter Example



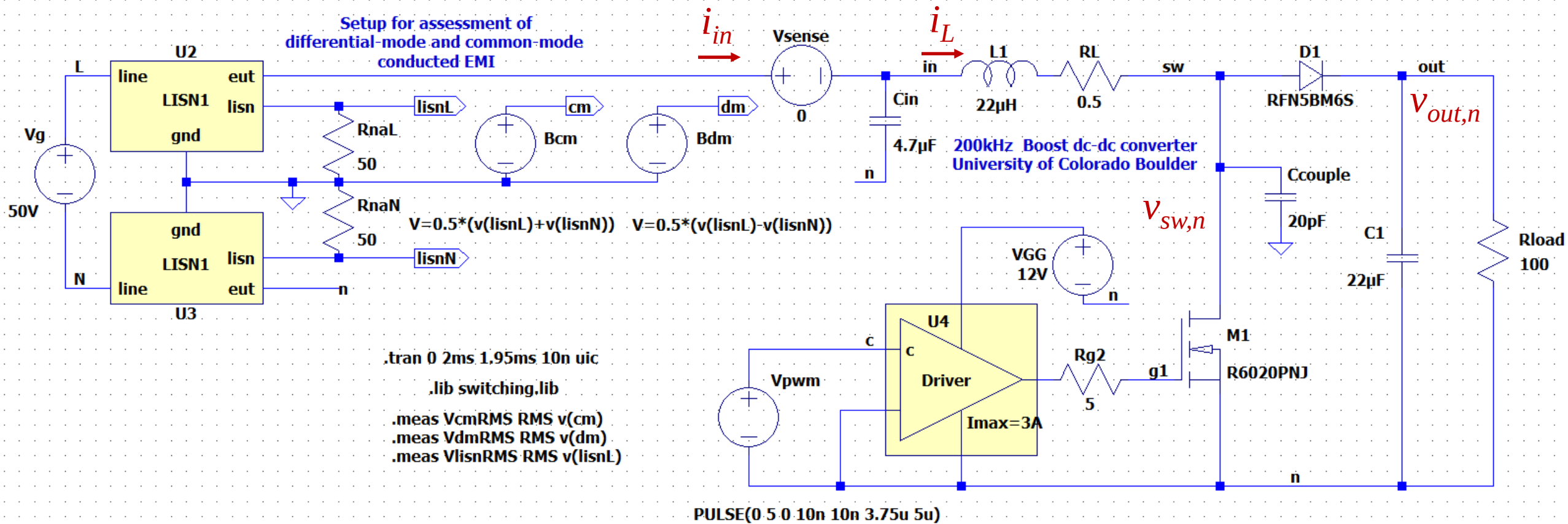
Input voltage $V_g = 50 \text{ V}$, duty cycle $D = 0.75$

Output voltage $V_{out} \approx 200 \text{ V}$, output power $P_{out} \approx 400 \text{ W}$

Switching frequency $f_s = 200 \text{ kHz}$



Conducted EMI



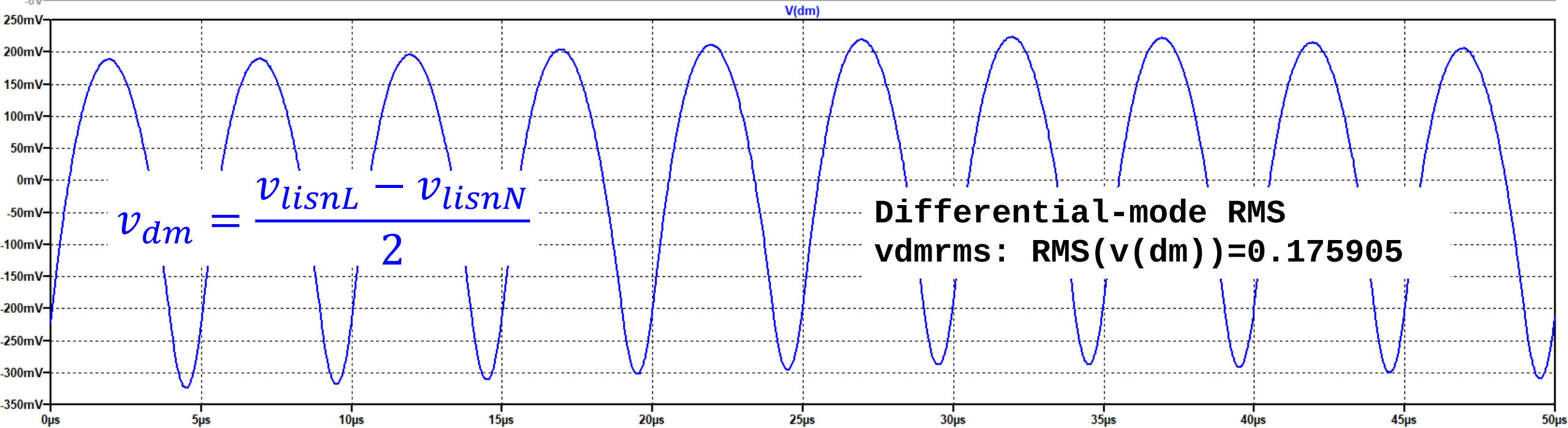
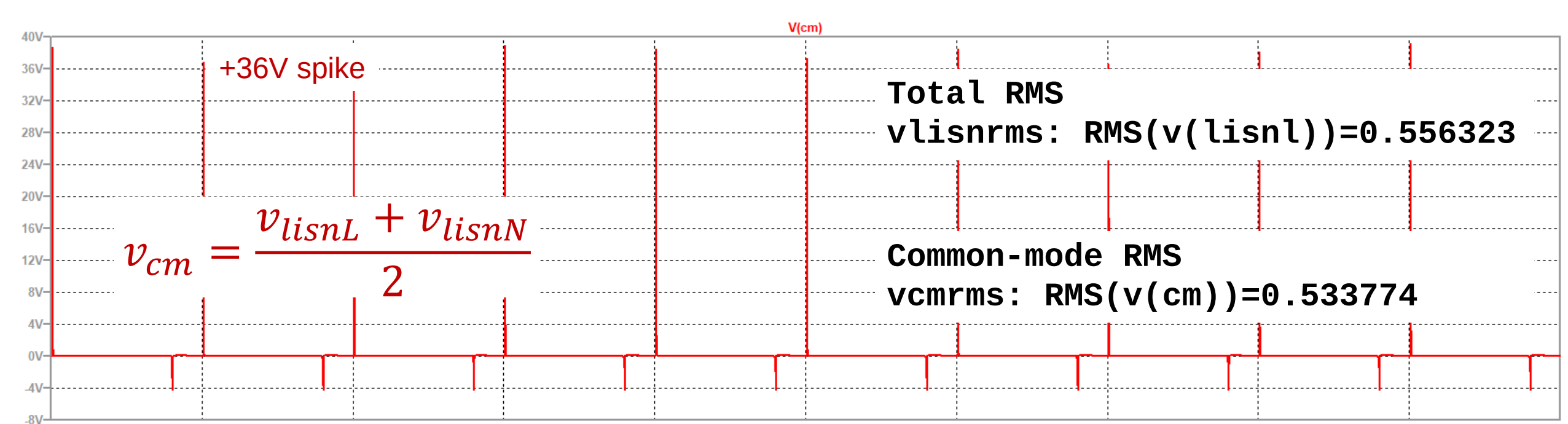
Overall EMI
measured by
the spectrum
analyzer

v_{lisnL}, v_{lisnN}

$$\left\{ \begin{array}{l} v_{dm} = \frac{v_{lisnL} - v_{lisnN}}{2} \\ v_{cm} = \frac{v_{lisnL} + v_{lisnN}}{2} \end{array} \right.$$

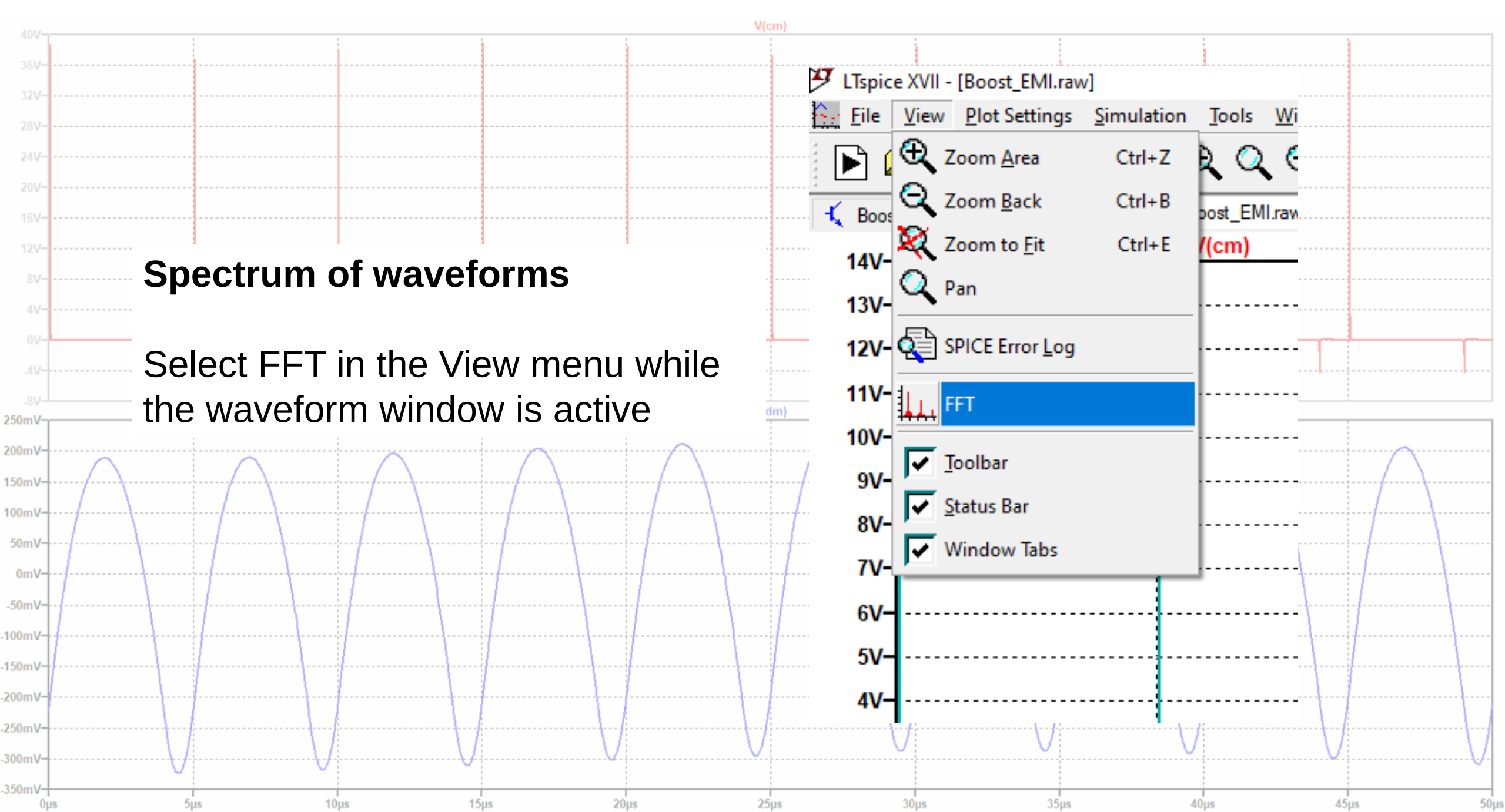
Differential-mode EMI

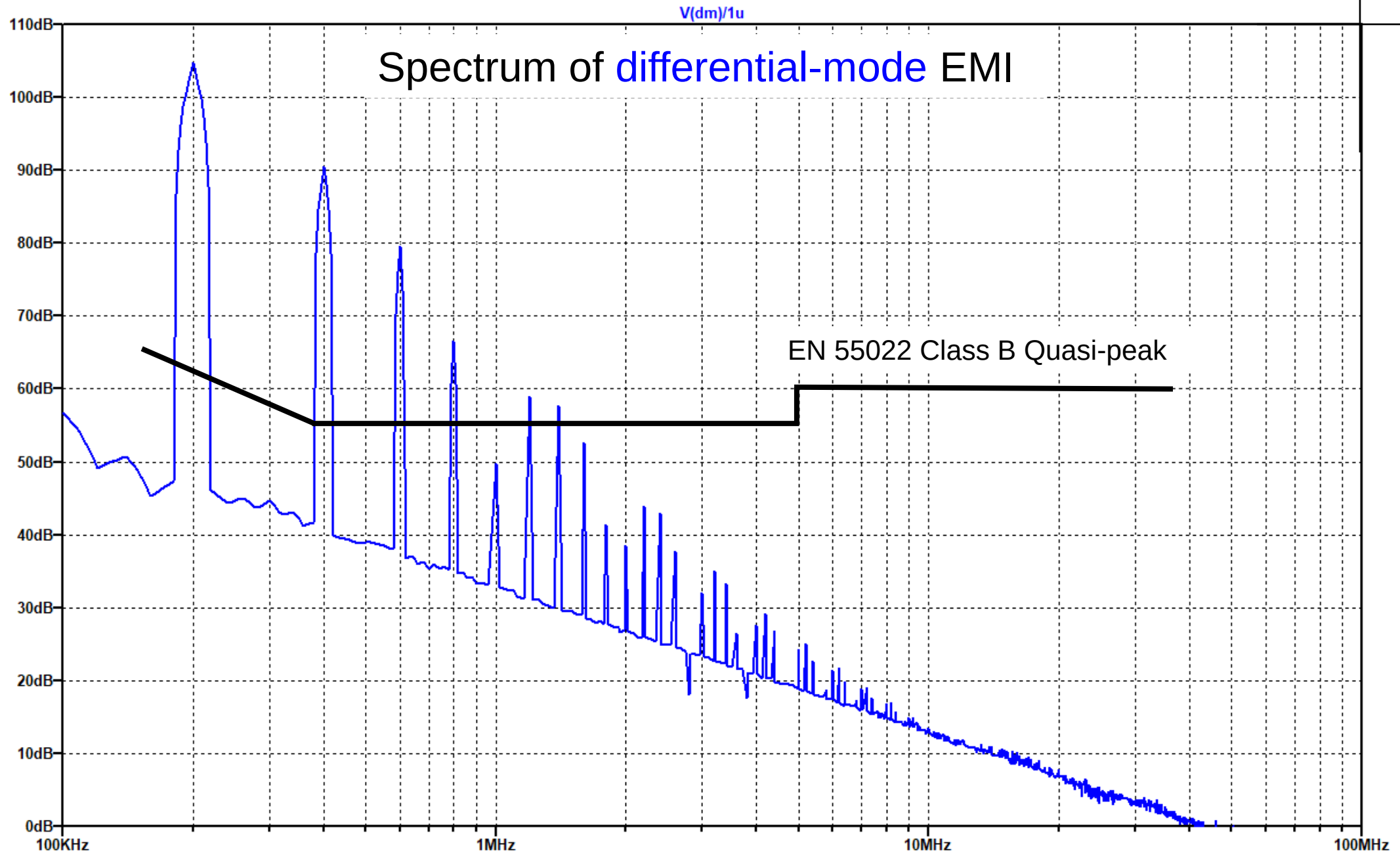
Common-mode EMI

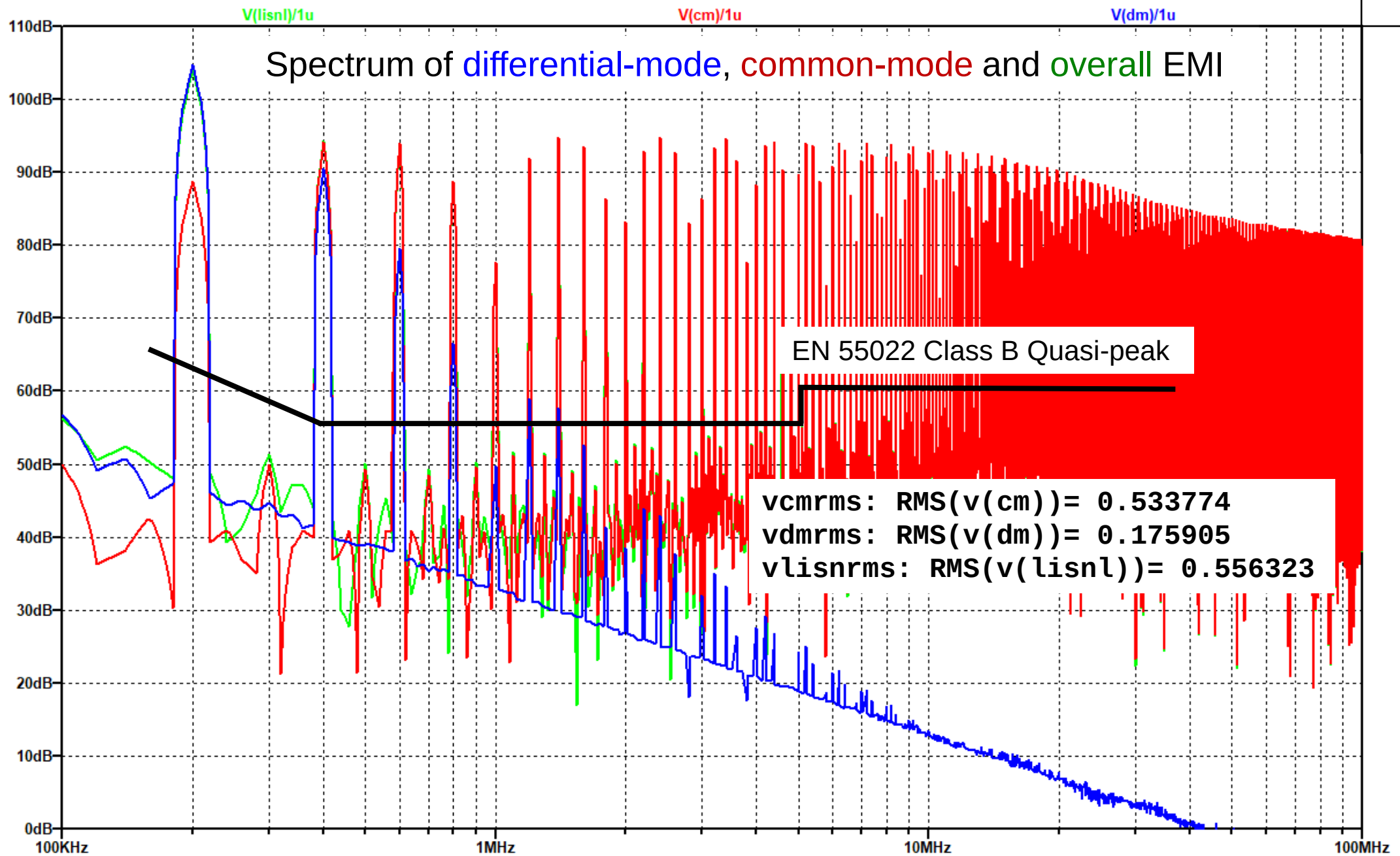


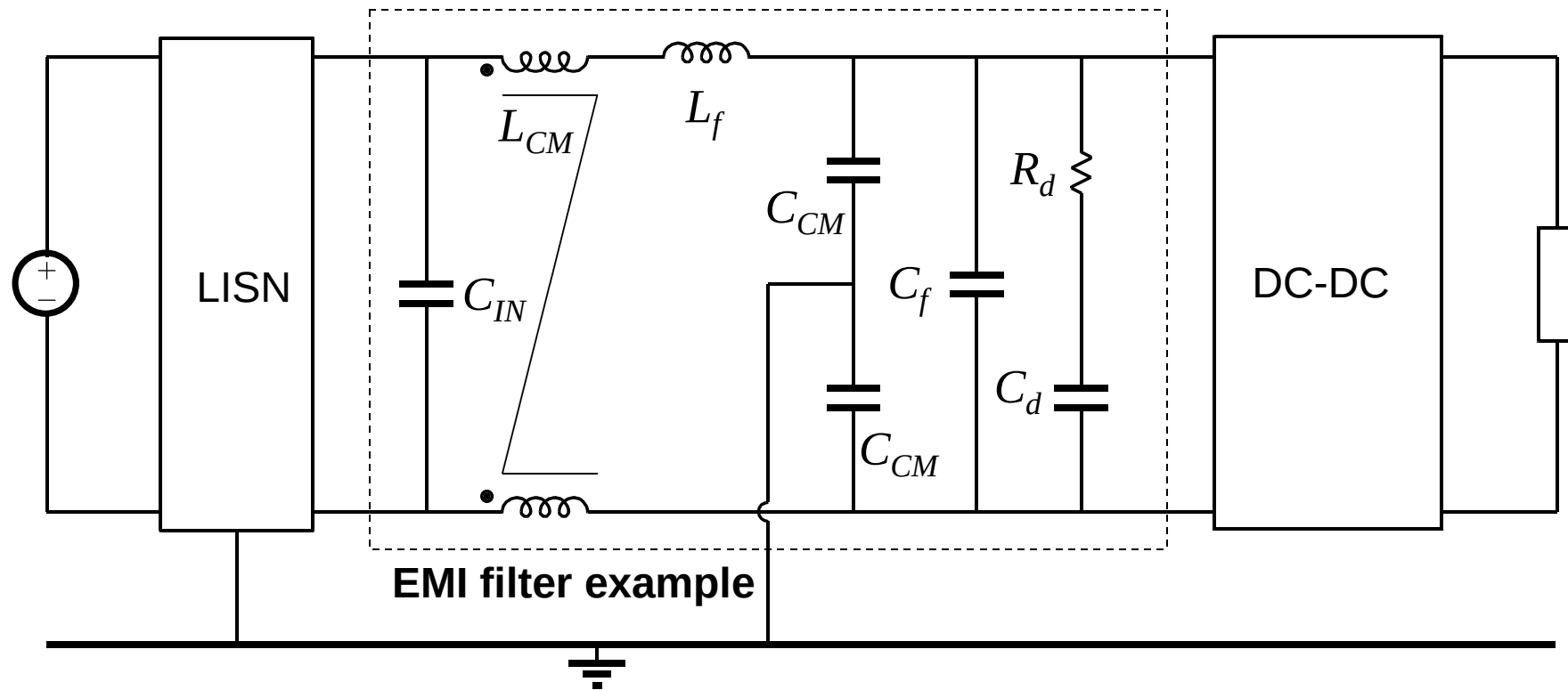
Spectrum of waveforms

Select FFT in the View menu while the waveform window is active

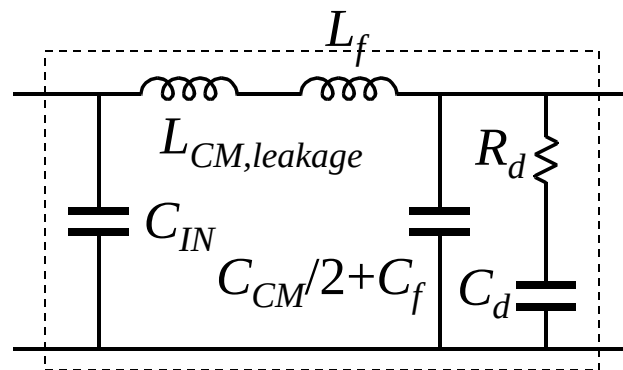




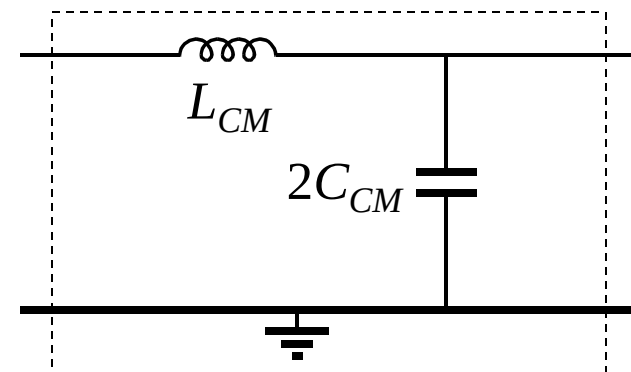




Differential-mode filter



Common-mode filter



Summary

- **Differential-Mode EMI** is determined by the converter input current ripple, which depends on converter configuration, parameters and switching frequency
 - Converters with pulsating input current generate larger differential-mode EMI
 - Filter damping is necessary to mitigate impact on converter frequency responses and closed-loop regulation performance
- **Common-Mode EMI** is determined by currents through parasitic capacitances between high dv/dt nodes and chassis ground, such as from transistor drain to heat sink
 - Common-mode EMI depends on dv/dt , component parasitics, circuit and mechanical layout
 - Electrical simulation models can be used to gain insights and compare CM performance, but are inadequate to evaluate or verify EMC compliance
- **Differential-mode** and **common-mode filters** are typically considered and designed separately